

Agent 07

Native / Rust Performance

Role: **Systems engineer**. The Rust hot-path crate — port scanner + filesystem walker. The 'we ported the slow part to Rust' layer: memory-safe, std-only, dependency-free.

Baseline health: **6.1 / 10**. Targets: Documentation 4→8, Test coverage 5→7.
Platform v2.3.2 • supervisor: main thread (senior dev)

Mission brief

Context: Agent 06 (Networking) bridges this crate's termaid-scan binary via backend/native.py — your JSON output is a contract it parses.

Objective: Document (/// doc-comments) → break down → harden the scanner + walker; write the JSON schema as the contract; add edge-case tests.

Boundaries: own native/ fully; the Tauri lib.rs native commands are a SHARED edge with Agent 09 — refine the command bodies, don't take the rest of desktop-mobile/.

Files owned

- native/Cargo.toml
- native/src/lib.rs — port scanner + pub mod fs
- native/src/fs.rs — directory walker
- native/src/main.rs — termaid-scan CLI
- native/src/bin/termaid-walk.rs — termaid-walk CLI
- native/tests/scan_test.rs, native/tests/fs_test.rs
- desktop-mobile/src-auri/src/lib.rs native commands — SHARED with Agent 09

Code is in the companion file Native_Rust_Code.md (attach both). Keep cargo build/test/clippy green — CI builds Rust.

Session flow

1. Orient — MASTER_INDEX, ARCHITECTURE, CODE_STYLE, RULES, LESSONS, BASELINE_HEALTH.
2. Kickoff brainstorm — today's ONE task + Definition of Done.
3. Directive 1 — /// doc-comments on every public item, Misfit.
4. Directive 2 — BREAKDOWN.md + JSON schema.
5. Directive 3 — Harden (scan bounds, edge cases).
6. Directive 4 — Health report vs baseline.
7. Hand-back — HANDOFF + files (.rs text) + INDEX + BREAKDOWN + health; append HISTORY; bump version.

Baseline health (this slice)

Category	Score	Note
Correctness	6	builds in CI; not built in sandbox
Security	7	memory-safe, std-only
Performance	8	the point: bounded thread pool
Architecture	7	clean, dependency-free
Tests	5	scan+fs tests; edges light
Documentation	4	needs /// doc-comments
Cohesion	6	JSON contract informal

Overall	6.1	fast safe core; under-documented
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History

Appended every session.

- 2026-06-14 · main · Kit created (baseline 6.1). Awaiting first session.